

LLD + DSA Prep Roadmap

Atin Sharma · SDE-1 Targets · 8-Week Plan · TUF+ LLD Course (71 topics)

■ LLD topic

■ DSA (daily, parallel)

■ Skip for SDE-1

Phase 1 Foundations + Creational Patterns Weeks 1–2 · ~14 days		
	LLD (45–60 min/day)	DSA (6–8:30 AM, unchanged)
Week 1 Days 1–7	TUF+ · Concepts (watch + notes) [Intro to LLD] [Software Design Principles] [SRP] [OCP] [LSP] [ISP] [DIP] [UML] [Class Diagrams] ■ After each SOLID lesson, write one TypeScript violation + one corrected version. 10 min max per principle.	Your existing TUF+ A2Z schedule. No change. DSA is non-negotiable before 8:30 AM.
Week 2 Days 8–14	TUF+ · Creational Patterns (watch + code) [Intro to Design Patterns] [Singleton] [Factory Method] [Builder Pattern] [Abstract Factory] [[SKIP] Prototype] ■ For each pattern: watch video → write from scratch in TS → write one real-world analogy in 2 lines. Prototype is low-priority skip.	Continue A2Z. Arrays / strings / two-pointer zone pairs well — Builder Pattern mirrors the step-by-step answer-building mental model.
Phase 2 Structural + Behavioural Patterns Weeks 3–4 · ~14 days		
	LLD (45–60 min/day)	DSA (6–8:30 AM, unchanged)
Week 3 Days 15–21	TUF+ · Structural Patterns [Adapter] [Decorator] [Facade] [Proxy] [[SKIP] Composite] [[SKIP] Bridge] [[SKIP] Flyweight] ■ Adapter, Decorator, Facade, Proxy are the SDE-1 core. Composite, Bridge, Flyweight are SDE-2 territory — skip entirely.	Continue A2Z. Sliding window / binary search zone — pattern-recognition skill transfers directly to design pattern identification.
Week 4 Days 22–28	TUF+ · Behavioural Patterns [Observer] [Strategy] [Command] [Iterator] [State] [[SKIP] Template Method] [[SKIP] Chain of Responsibility] [[SKIP] Visitor] [[SKIP] Mediator] [[SKIP] Memento] ■ Observer + Strategy + Command = holy trinity. State is needed for Vending Machine + ATM. Iterator is lightweight — 30 min max. Skip the rest.	Continue A2Z. Linked lists / stacks / queues — Command pattern is literally an operation queue under the hood.
Phase 3 TUF+ Advanced Topics (selective) Week 5 · 7 days		
	LLD (45–60 min/day)	DSA (6–8:30 AM, unchanged)
Week 5 Days 29–35	TUF+ · Advanced (selective reads) [Dependency Injection] [Exception Handling (LLD)] [Building Resilient Systems] [Best Practices in LLD] [All About APIs] [[SKIP] Multithreading + Concurrency] [[SKIP] Thread Pools] [[SKIP] Thread Safety] [[SKIP] Locks + Deadlocks] [[SKIP] Producer Consumer] [[SKIP] Database Design] ■ Multithreading is SDE-2+. At SDE-1 you get at most 1 conceptual question. DI + Error Handling + Best Practices + API design are 1-day reads each.	Continue A2Z. If in tree / graph zone this week, cut LLD to 30 min on heavy graph BFS/DFS days — protect DSA time.

Phase 4 Interview Problems — Full Design + Code Weeks 6–8 · ~21 days		
	LLD (1 problem per 2–3 days)	DSA (6–8:30 AM, unchanged)
Week 6 Days 36–42	TUF+ Part-1 · Tier 1 problems [Parking Lot Design + Code] [Vending Machine Design + Code] [Logging Framework Design + Code] ■ Protocol: Read requirements → close TUF+ → draw class diagram in 15 min → write TS code in 30 min → compare → re-do from scratch after 2 days.	Continue A2Z. DSA load is normalizing now. No cuts to DSA time.
Week 7 Days 43–49	TUF+ Part-2 · Tier 1–2 problems [ATM Machine Design + Code] [Task Management System Design + Code] [PubSub System Design + Code] ■ PubSub maps directly to your BullMQ + Upstash work at OpenBiz. ATM uses State pattern — explicitly call it out during the design step.	Continue A2Z. If you hit DP here, it is the heaviest DSA week. Reduce LLD to 1 problem this week if needed — protect DSA time.
Week 8 Days 50–56	TUF+ Part-3 · Tier 2–3 problems [Elevator System Design + Code] [Ride Booking App Design + Code] [Hotel Management System Design + Code] [Traffic Signal System Design + Code] ■ Ride Booking is your strongest problem — directly analogous to VyaparGPT routing. Use it as your anchor story in interviews.	Continue A2Z. Mock interview week — schedule 1 timed LLD mock (45 min) with your Discord study partner.

Full Skip List — Topics to bypass at SDE-1 bar

Topics from TUF+ LLD to skip entirely at SDE-1 bar

Prototype Pattern Composite Pattern Bridge Pattern Flyweight Pattern Template Method Chain of Responsibility Visitor Pattern Mediator Pattern Memento Pattern Multithreading + Concurrency (full module) Thread Pools + Executors Thread Safety + Synchronization Locks + Deadlocks Producer Consumer Database Design + Integration Digital Wallet (Part 3) Music Streaming Platform Streaming Protocols

These are SDE-2+ or niche topics. Revisit only after you have SDE-1 offers in hand.

Drill Protocol for Every Tier 1 Problem

Step	Action	Time
1	Read problem requirements on TUF+	5 min
2	Close TUF+. Draw class diagram from memory	15 min
3	Write TypeScript classes from diagram	30 min
4	Reopen TUF+, compare with solution, note gaps	10 min
5	Re-do from scratch after 2 days without notes	45 min

Time Budget Summary

	LLD Time/Day	LLD Mode	DSA	Note
Wks 1–2	45–60 min	Watch + write code	6–8:30 AM	Concepts phase — no compromise on DSA
Wks 3–4	45–60 min	Watch + write code	6–8:30 AM	Pattern phase — heaviest LLD learning load
Week 5	30–45 min	Read + notes only	6–8:30 AM	Cut to 30 min if graph zone in A2Z
Wks 6–8	60–90 min	Design + full code	6–8:30 AM	Cut to 1 problem/week if DP zone in A2Z